PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

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Abstract

Segmentation models in medical imaging are trained on one acquisition setting and must generalize to others. Label-generative domain randomization, of which SynthSeg is the archetype, addresses this by discarding the acquired scan and resampling intensities from a per-label Gaussian mixture over a dense *anatomical label map – an annotation far richer than the segmentation task itself provides, and one that does not preserve texture: real gradients and partial-volume detail become piecewise noise. We ask when that loss costs accuracy, and find the answer is a property of the* task. *Where the target is separated from its surroundings by an anatomical interface – encapsulated abdominal organs, bounded by fat planes – destroying texture costs nothing. Where no interface exists – diffuse glioma, infiltrating beyond both the enhancing margin and the peritumoral FLAIR abnormality – the target must be inferred from tissue appearance, and destroying texture is expensive. We exploit this with PALETTE, a single-source augmentation needing no dense label map: it partitions a scan's foreground by unsupervised k-means, sub-divides each class spatially by a Voronoi partition, and applies a signed random affine remap to the* real *intensities of each sub-region. Remapping existing intensities rather than replacing them leaves edges, gradients and partial-volume detail largely intact; only the contrast relationships between regions are randomized. A controlled ablation isolates this: holding the partition and per-region statistics fixed and swapping only the real-intensity remap for label-style noise fill costs 7.7 and 7.2 Dice on tasks with no tissue interface, against $+1.1$ and -1.1 on organ tasks with one, and -4.5 and -1.4 on a third, far more densely-labelled interface task (31-region healthy-brain parcellation) whose regions are, on inspection, tissue-interface targets too – negative rather than merely small, plausibly because so much more of the volume is relabelled at once, but never a significant gain either way. Across cross-contrast, cross-modality and cross-dataset transfer on healthy-brain, MS-lesion, brain-tumour, abdominal-organ, and liver-tumour segmentation, one nnU-Net configuration trained with PALETTE attains the best Dice on every task and improves significantly over all five competing methods on Dice, remaining at or near the best on boundary distance without being the outright aggregate winner there.*

1. Introduction

Medical image segmentation models are trained on one acquisition setting and are frequently applied to another: a different scanner, sequence, or imaging modality than the one seen during training. MRI contrast is a protocol choice rather than a physical constant, so the same anatomy yields incompatible intensity distributions across sequences, and a network that has only seen T1-weighted brain images can fail outright on T2-weighted or FLAIR images of the same anatomy [2, 29]. Acquiring and annotating training data for every contrast a model might encounter is not practical: manual segmentation is expensive, and clinical sites rarely share raw data across institutions [26, 43]. The single-source domain generalization setting — train on one labelled dataset in one contrast, generalize to unseen contrasts and modalities without any target-domain data — is therefore the setting that matters for deployment, and the one a growing body of work addresses [31, 38, 46, 48].

The dominant single-source strategy is label-generative domain randomization, popularized by SynthSeg [2]. Given a *dense* anatomical label map — a voxel-wise parcellation of the whole image, not the sparse structure-of-interest annotation a segmentation dataset normally provides — intensities are resampled from a per-label Gaussian mixture, producing images of arbitrary, unrealistic contrast that force the network to ignore contrast and rely on shape and layout [2]. This is effective, but carries two structural costs. First, it is *label-driven*: the dense parcellation it resamples from is a richer annotation than the task itself requires, so generating synthetic contrast costs exactly the expensive labelling the method aims to reduce reliance on. Second, and less discussed, it *does not preserve texture*: replacing every labelled region with piecewise Gaussian noise discards the real gradients, partial-volume boundaries, and unlabelled fine structure the acquired scan contains. Layout survives; texture does not.

We ask whether discarding texture is necessary for contrast robustness, or merely the path of least resistance once a dense label map is assumed. Our central finding is that the answer depends on the *task*: preserving real intensity structure matters where the segmentation target is separated from its surroundings by a genuine anatomical interface, and not where it is not. That texture carries diagnostic signal in exactly the tasks where no such interface exists is established independently of us — texture features discriminate MS lesions from normal-appearing white matter [47], and diffuse glioma infiltrates beyond every margin conventional MRI shows [12, 40] — and our contribution is to show that this distinction governs whether a texture-preserving augmentation helps at all.

We exploit it with **PALETTE**, a single-source augmentation that needs no dense label map. PALETTE partitions the foreground of a scan into intensity classes with unsupervised 1-D k-means, further subdivides each class into spatial sub-regions with a Voronoi partition, and applies a signed random affine remap to the *real* intensities within each sub-region. Because the transform remaps existing intensities rather than replacing them with sampled noise, edges, gradients, and partial-volume detail largely survive within each sub-region; only the region-to-region contrast relationship is randomized, including sign inversions no physical acquisition protocol produces. The optional final step is the only one that uses labels, and it uses only the sparse task labels already present in any segmentation dataset — never a dense parcellation.

We test the mechanism claim directly rather than inferring it from aggregate scores. Holding the k-means/Voronoi partition and the per-region statistics fixed, we replace only the real-intensity affine remap with label-style noise fill, matching SynthSeg's resampling inside PALETTE's own geometry. The swap costs 7.7 and 7.2 Dice on tasks whose targets have no tissue interface, against $+1.1$ and -1.1 on organ tasks that do — a separation by target type, not a uniform offset — and $-4.5/-1.4$ on a third, far more densely-labelled interface task whose regions turn out, on inspection, to be tissue-interface targets too, just packed across near-total brain volume (§4.3). An independent gradient-orientation measurement confirms that texture is what changes at the fill-swap step.

Empirically, training a single nnU-Net [16] configuration — fixed across datasets, with only the epoch budget varying — we evaluate cross-contrast, cross-modality, and cross-dataset transfer on five segmentation tasks: healthy brain, MS lesion, brain tumour, abdominal organ, and liver tumour. We compare against a no-augmentation baseline, both variants of SynthSeg (the original dense-label-driven variant, and an EM variant with no public implementation that we re-implement), SRCSM [41], and Auglab [29], a published augmentation pipeline designed for cross-modality training and the strongest competitor. PALETTE improves significantly over every competing method on Dice at the task level, and remains at or near the best on boundary distance without always being the outright aggregate winner there. No competitor is reliably second: the runner-up changes identity across settings, so PALETTE is the only method at or near the top throughout.

In summary, we make the following contributions:

- **A task-level account of when texture preservation matters.** Preserving real intensity structure helps where the segmentation target has no anatomical interface separating it from its surroundings, and not where it has one. We state this as a prediction and test it with a controlled dissociation rather than asserting it post hoc.
- **A causal ablation isolating texture.** Holding the intensity and spatial partition and the per-region statistics fixed and swapping only the real-intensity remap for label-style noise fill separates our headline tasks by target type, a separation a third, far more densely-labelled interface task also follows, just with a larger negative swing. An independent gradient-orientation measurement confirms that texture, not the partitioning scheme, is what changes at that step.
- **PALETTE**, the single-source contrast augmentation that exploits this, requiring no dense anatomical label map and using only the sparse task labels a segmentation dataset already provides.
- **Breadth evidence.** Across cross-contrast, cross-modality and cross-dataset transfer on healthy-brain, MS-lesion, brain-tumour, abdominal-organ, and liver-tumour segmentation, with one configuration reused across datasets, PALETTE attains the best Dice on every task and improves significantly over all five competing methods on Dice, remaining at or near the best on boundary distance without always being the outright aggregate winner there.

2. Related Work

2.1. Domain generalization for medical segmentation

Domain generalization (DG) targets unseen distributions without access to target data, unlike unsupervised domain adaptation, which assumes unlabelled target images and solves an easier problem [37]. Most medical DG methods are *multi-source*, interpolating across several labelled domains by meta-learning, feature alignment, frequency-space exchange, multi-frequency architectures, foundation-model scale, or diffusion-based stylization [1, 24, 26, 30, 35, 43] – all of which still rely on source diversity. We instead work in the *single-source* regime — one labelled dataset, no target data, no second source – the setting that matters when target-domain data is unavailable at training time, which is routine in clinical deployment [31, 48]. This rules out cross-domain interpolation and makes architecture-coupled methods orthogonal to our contribution: our mechanism acts at the input level and leaves the segmentation network a fixed nnU-Net [16].

2.2. Synthesis-based contrast-agnostic training

The dominant single-source strategy is label-generative domain randomization, introduced by SynthSeg [2]: intensities are sampled from a per-label Gaussian mixture over an anatomical segmentation, yielding images of arbitrary contrast that force contrast-invariance. This family carries two structural costs: it is *label-driven* (a dense anatomical label map is required per subject – SIAM shifts the dependence from label *quantity* to *quality* with curated templates, but each still needs a full dense parcellation [42]), and it is *texture-destroying*, since per-label Gaussian sampling replaces real gradients, partial-volume detail and unlabelled structure with piecewise noise, retaining layout but not texture. The lineage has been specialized to white-matter and MS lesions and to atlas-based expectation-maximization [3, 4, 22]. Among SynthSeg's own variants we take the original (*noEM*), which resamples intensities from a dense anatomical label map and never looks at the acquired scan, and the *EM* variant, which instead fits its per-class intensity model to the real image by expectation-maximization and so needs only the sparse task labels a segmentation dataset already provides, not a dense parcellation. The EM variant has no released implementation; ours follows the mechanism described in the original paper [2], and a faithful re-implementation is itself a contribution.

2.3. Texture as a segmentation signal

That local texture, not only intensity level, carries information a segmentation model can use is established independently of us: second-order gray-level statistics discriminate image content beyond simple intensity [13], and quantitative texture features from tumour regions carry prognostic signal across CT, MRI and PET [21]. Closer to our setting, Kang et al. disentangle structure from texture in cross-domain image translation with a texture co-occurrence loss, targeting the same failure mode we do — naive translation distorts fine texture and degrades downstream segmentation [18]; a texture-fidelity loss has likewise been added to GAN-based lesion augmentation [10]. That some targets resist boundary-based methods for lack of a sharp edge, and are better served by texture-based segmentation, has been argued directly for diffuse white-matter lesions [20]. None of this prior work, to our knowledge, connects texture preservation in an *augmentation* to *which* segmentation targets it helps; that connection, tested causally, is our contribution.

2.4. Single-source intensity and appearance augmentation

Where the first line of single-source methods synthesizes from a label map, a second manufactures appearance diversity directly from the real image, with no generative label model. General-purpose libraries stack intensity, spatial and artifact transforms on an nnU-Net trainer; the recent Auglab pipeline [16, 29] is one such GPU-accelerated framework and is the strongest competitor in our experiments, and BigAug stacks a comparable set of coarse intensity, quality and spatial perturbations under the same single-source DG protocol we use [48]. Our method inserts the PALETTE transform as one additional operator inside the Auglab pipeline, so the comparison against Auglab directly isolates PALETTE's contribution. Closer to our operator are random-filter methods, which vary appearance with an untrained nonlinear filter: GIN applies randomly re-initialized convolutions for label-free appearance variants [31] (the GIN-alone configuration we compare against is an ablation in the original paper of a two-view objective on 2-D slices, so our comparison is illustrative of the random-filter family rather than a reproduction of that paper's proposed method); related variants establish the random-filter principle for natural images, perturb intensities per class using the task mask, add learned diffusion generators, or perturb the Fourier amplitude spectrum instead of the image itself [5, 25, 33, 38, 46, 49]. Our operator instead partitions the foreground into intensity classes and Voronoi subregions and applies a signed *affine* remap of the real intensities within each, invertible and gradient-preserving by construction. The closest peer is SRCSM [41] (semantic-aware random convolution with source matching), which also localizes its perturbation region-wise in true 3-D and, like PALETTE, layers exactly one label-consuming component over an otherwise label-free step: at training time it draws a fresh untrained convolution per anatomical class from the ground-truth map (needing a Gaussian smoothing pass to control the boundary-contrast artifacts this introduces), and at test time applies a label-free histogram matching onto the source distribution. The difference from PALETTE is not label use versus none, but *how* that step transforms the signal: an untrained nonlinear convolution versus a deterministic affine remap that leaves relative intensity structure inside each region intact.

2.5. Cross-contrast evaluation protocol

We evaluate a model trained on a single real input modality on every held-out modality of the *same* segmentation task, including the training modality itself, across four tasks spanning brain, abdominal and tumour anatomy in MRI and

CT; a sixth task, spine segmentation, is a planned extension. This single-sequence-in, every-sequence-out design is not unprecedented – Auglab's own evaluation follows the identical protocol for spine segmentation [29] – and our contribution is breadth (four distinct cross-contrast tasks, not one) and, for two of them, a further axis: MS-lesion and abdominal-organ segmentation are each also evaluated against independent external test datasets acquired at different sites, so the training dataset's own test set is one point among several rather than the only held-out evidence. A fifth task, liver-tumour segmentation, trains on a single modality and so has no cross-contrast axis of its own, but is evaluated against two independent external cohorts the same way, extending this same-site-is-not-enough argument to a setting where cross-contrast transfer cannot be measured at all.

3. Method

We describe PALETTE (§3.1), how it is applied during training (§3.2), and the causal ablation used to isolate texture preservation as the source of its effect (§3.3).

3.1. The PALETTE transform

Let x denote the intensity volume of a training scan, min-max normalised to $[0, 1]$, and $F \subset \Omega$ its foreground voxels (an intensity threshold; $x_v > 0.01$ in our implementation). PALETTE applies four steps at every training iteration, illustrated end-to-end in Fig. 1 on one real CHAOS (abdominal MRI) and one real OnHarmony (brain MRI) volume. The figure's five panels (a)–(e) are deliberately lettered rather than numbered 1–5 so they are never confused with the four method steps below: panels are *intermediate diagnostic renderings*, one per step plus the input, not a one-to-one visual walkthrough of the steps themselves (panel (e), the final training input, is the only panel that shows the texture the transform actually produces; (b)–(d) show flat preview values, defined in the caption).

1. Intensity partition. We run 1-D k-means on the foreground intensities $\{x_v : v \in F\}$ with k sampled per iteration from $\{2, \ldots, 6\}$, producing k intensity classes $C_1, \ldots, C_k$ (Fig. 1, panel (b)). This partition is unsupervised and requires no anatomical label map: it groups voxels by intensity alone, so the same operator applies unmodified to any input image regardless of its modality or contrast — CT, T1, T2, FLAIR, or otherwise. This step is occasionally skipped in favour of a single global remap (step 3 below, with $C = 1$); per-step probabilities are set once and never tuned per dataset, and are listed in Sec. 7.

2. Spatial sub-parcellation. Each intensity class C_i is further subdivided into spatial sub-regions by a Voronoi partition seeded with points sampled within C_i (Fig. 1, panel (c) shows the combined effect of this step and the remap of step 3), giving sub-regions $R_{i,1}, \ldots, R_{i,m_i}$ that are contiguous in space as well as similar in intensity. This second partition breaks the spatial correlation of a purely intensity-based grouping: two voxels with the same intensity in different parts of the anatomy can land in different sub-regions and receive independent contrast perturbations. Each class is usually, but not always, sub-split this way (Sec. 7).

3. Signed affine remap. For each sub-region R, let $\bar{x}_R$ denote its mean intensity, $\mu_R \sim U(0, 1)$ a resampled target mean, and draw $\alpha_R \sim \pm U(0.5, 2.0)$ (a magnitude in $[0.5, 2.0]$ with an independently sampled sign). Every voxel $v \in R$ is remapped as

$$y_v = \mu_R + \alpha_R \, (x_v - \bar{x}_R). \tag{1}$$

Because Eq. (1) is a per-region affine function of the *real* intensity x_v, it preserves the relative structure of the signal within each region: edges, gradients, and partial-volume transitions between adjacent voxels are scaled, not replaced. Only the region-to-region contrast relationship is randomized, and $\alpha_R < 0$ additionally allows local contrast inversions that no acquisition protocol produces on its own, widening the space of contrasts the network sees beyond what any single real sequence could supply. Background voxels are zeroed at this step (and remain zero through the rest of the pipeline). The result is occasionally Gaussian-blurred, mostly not (Sec. 7).

4. Anatomical-label affine remap (optional). Steps 1–3 are unsupervised, so the k-means/Voronoi partition need not respect a true anatomical boundary: when two adjacent tissues share similar intensity, step 3 can by chance assign them the same target contrast and erase that boundary rather than randomizing it. This step corrects for that: for each real anatomical label ℓ present, we resample a fresh (μ_ℓ, α_ℓ) and apply Eq. (1) *again*, restricted to ℓ's voxels, **on top of the step-3 output** rather than the original voxels (Fig. 1, panel (d)) – so it composes with, rather than replaces, the unsupervised remap, and ℓ's voxels still inherit the internal structure steps 1–3 imposed on them. Because label assignment is independent of the unsupervised partition, this reintroduces a full-strength contrast discontinuity at the true label border regardless of whether step 3 preserved it, sharpening the boundary cue available to the network – not the underlying anatomy. It consumes only the labels the segmentation task already requires, so it needs no additional annotation, and is optional: disabling it (`label_remap_prob`= 0) degrades PALETTE gracefully to steps 1–3 alone, e.g. on an unlabelled dataset. A final foreground z-score normalisation follows an optional second blur pass, giving the training input shown in panel (e) of Fig. 1. Per-step probabilities are set once, never tuned per dataset, and listed in full in Sec. 7.

3.2. Training protocol

PALETTE is applied on-the-fly as one contrast-augmentation operator inside an existing GPU-accelerated nnU-Net augmentation pipeline [16, 29], alongside that pipeline's standard spatial and artifact augmentations; it adds no learned parameters and no measurable training-time overhead. A single configuration – the same k-range, sub-region count, blur schedule, and α range given in §3.1 – is tuned once and reused unchanged across every dataset and modality in Sec. 4; no dataset receives its own hyperparameter sweep.

3.3. Causal ablation: isolating texture preservation

PALETTE differs from label-generative synthesis (e.g. SynthSeg) along two axes at once: the partition it perturbs (unsupervised k-means + Voronoi vs. a supplied anatomical label map as the sole generative substrate) and the perturbation itself (an affine remap of real intensities vs. resampling from a per-label Gaussian mixture). Comparing PALETTE to SynthSeg directly therefore cannot attribute an advantage to either axis in isolation.

We construct a controlled variant, *noise-fill*, that holds the k-means/Voronoi partition and the per-region statistics (μ_R, α_R) from Eq. (1) exactly fixed, but fills each sub-region with samples drawn from a Gaussian with that region's $(\mu_R, \alpha_R \sigma_R)$ instead of remapping the real intensities inside it – i.e. the same partitioning scheme, with SynthSeg-style noise substituted for the affine remap. Because the partition and target statistics are identical to PALETTE's own, any difference in downstream performance between PALETTE and noise-fill isolates the one variable that changed: whether real texture is preserved or discarded within each region. We report this comparison separately for structures with no tissue interface (fine internal gradients, low boundary contrast) and structures bounded by one (identified by a sharp, high-contrast border) in Sec. 4, as the texture-preservation hypothesis predicts a widening gap specifically on the former.

4. Experiments

4.1. Setup

Protocol. We train a single real-intensity source modality per dataset and evaluate on held-out modalities of the *same* segmentation task, using one fixed nnU-Net [16] configuration across all settings. No target-domain data, of any modality, is seen during training. Our full method, denoted **Ours** throughout, is PALETTE applied as one additional contrast-augmentation operator inside the Auglab pipeline [29]; the *Auglab* baseline is the same pipeline with the PALETTE operator removed, so the two differ by exactly the PALETTE transform. Every method is trained with the identical nnU-Net schedule — SGD with Nesterov momentum (0.99), the nnU-Net `polyLR` decay from an initial rate of 10^{-2}, 250 mini-batches per epoch, and the combined Dice-plus-cross-entropy loss — and differs only in the augmentation applied at the input. The epoch budget is fixed per dataset and shared by every method on that dataset (2500 for Brats-GLI, 2000 for ON-Harmony, Open-MS, and Liver-HCC, 200 for CHAOS), and all results are 3-fold cross-validated, as is standard practice in medical image segmentation and so that every comparison rests on per-case statistics rather than a single split. All significance tests are paired per case and Holm-corrected [15]; full test construction, configuration details, and per-dataset descriptions (subject counts, acquisition protocols, and label provenance) are given in Secs. 7 to 9.

Datasets. We evaluate on five tasks drawn from nine datasets (Tab. 1), spanning abdominal CT/MRI, brain MRI, brain tumor MRI, and liver MRI. Four tasks (Brats-GLI, CHAOS, Open-MS, ON-Harmony) train on a single source modality and are tested on every other contrast of that task *and* the training contrast itself – eight source-modality settings in total; no target modality is seen during training. CHAOS [19] additionally transfers cross-modality to held-out CT from AMOS [17] and SLIVER07 [14]; ON-Harmony [44] reference labels are silver-standard, obtained on T1w with FreeSurfer [7] and registered to the other contrasts. The fifth task, Liver-HCC (hepatocellular carcinoma tumour segmentation on ATLAS-Liver-HCC [34]), trains on a single modality (contrast-enhanced T1w) with no second in-house contrast to cross-evaluate against; its generalization axis is cross-*dataset* rather than cross-*contrast* – the same checkpoints are additionally tested on two independent external cohorts, one same-contrast-family (Liver-HccSeg [9]) and one genuinely cross-contrast (T2WI/DWI, the HCC subset of LLD-MMRI [27], with masks produced using MedSAM2 [28]). It therefore contributes to the task-level comparison of Tab. 2 but not to the per-setting breakdown of Sec. 9 or the causal-ablation ladder of Sec. 4.3, both of which require the cross-contrast axis the other four tasks have and this one does not.

Baselines. SynthSeg comes in two variants, which we refer to throughout by how they obtain their per-class intensity model: *noEM*, the released variant, which reads it off a dense anatomical label map; and *EM*, which estimates it from the image by expectation-maximization and therefore needs no dense parcellation. We compare against five methods. (1) *No-augmentation*, the nnU-Net default training recipe. (2) *SynthSeg-noEM* [2], the original label-generative variant (the implementation released by the authors), which

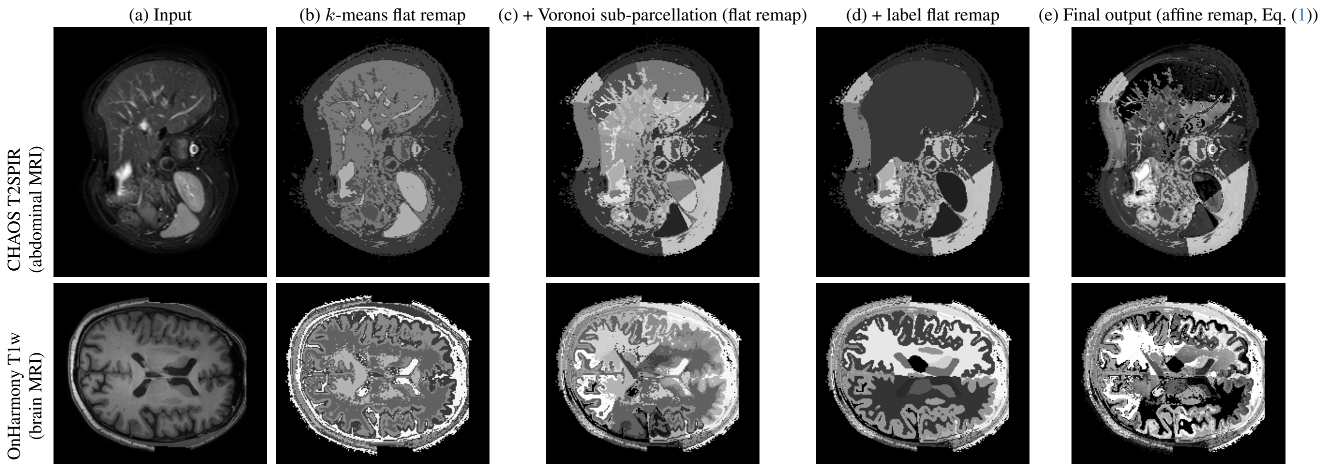

Figure 1. The PALETTE pipeline on one real CHAOS T2SPIR volume (top) and one real OnHarmony T1w volume (bottom); every panel is a deterministic function of the same real volume (no external model), min–max normalised to $[0, 1]$ for display only. Panels (b)–(d) paint each region, sub-region, or label with its precomputed target mean μ from Eq. (1), isolating that flat value from the texture applied on top of it; panel (e) is the actual training input, produced by applying the full signed affine remap of Eq. (1) with real texture (not a flat fill) at both the region and, where available, anatomical-label level.

Task	Dataset	Train modalities	Test modalities
Multi-Organ	CHAOS	T1-in, T2-SPIR	T1-in, T1-out, T2-SPIR, CT
	AMOS	–	CT, MRI
	SLIVER07	–	CT
Brain Glioma	Brats2024-GLI	T1n, T2w	T1c, T1n, T2w, T2-FLAIR
Brain MS	Open-MS	FLAIR, T1w	FLAIR, T1w, T2w
Healthy Brain	ON-Harmony	T1w, T2w	T1w, T2w, BOLD, DWI, EPI, GRE
Liver Cancer	ATLAS-Liver-HCC	CE-T1w	CE-T1w
	LLD-MMRI-HCC	–	T2WI, DWI
	LiverHccSeg	–	CE-T1w (4 phases)

Table 1. Per-task dataset roster. AMOS, SLIVER07, LLD-MMRI-HCC, and LiverHccSeg are cross-dataset transfer targets only ("–": never trained on). Liver Cancer trains on a single modality, so its generalization axis is cross-dataset rather than cross-contrast (Sec. 4.1). Subject counts, acquisition protocols, and label provenance are in Sec. 8.

	Brats-GLI	CHAOS	ON-Harm.	Open-MS	Liver-HCC	**overall**	p vs ours
Baseline	21.7	29.3	31.1	21.0	42.0	29.0	1.0e-46
SynthSeg-noEM	8.6	59.6	66.3	4.3	19.0	31.6	5.3e-74
SynthSeg-EM	43.6	83.0	67.4	33.9	47.3	55.1	5.2e-12
SRCSM	42.1	83.9	66.6	33.0	41.2	53.3	1.1e-32
Auglab	46.7	83.3	67.4	34.3	46.9	55.7	3.4e-10
PALETTE (ours)	**46.8**	**85.0**	**68.2**	**38.5**	**49.7**	**57.6**	—

Table 2. Task-level Dice (%). Each column pools that task's held-out contrasts and both source modalities, plus (CHAOS, Liver-HCC) an external cross-dataset test set pooled by contrast group so that a modality tested by several external sources does not outvote one tested by fewer (Sec. 9); **overall** weights tasks equally. Liver-HCC trains on a single modality (Sec. 4.1), so its column is cross-dataset rather than cross-contrast evidence. HD95, a secondary metric here, is reported separately in Tab. 7. p: Holm-corrected one-sided sign-flip test on the macro-averaged per-case difference, stratified by task. PALETTE attains the best Dice on every task and improves significantly over every competitor.

synthesises each training image from a dense anatomical label map alone, ignoring the acquired scan entirely. (3) *SynthSeg-EM*, the variant described in the original work but never released publicly, which we re-implement; unlike SynthSeg-noEM it does not require a dense anatomical label map, but it is not dense-label-free either — it is compatible with and relies on the sparse task labels naturally available in a segmentation dataset, which it uses to estimate its per-class intensity model. (4) *SRCSM*, a semantic random-convolution contrast-synthesis augmentation, included as a second strong synthesis-based competitor that is neither label-generative nor a member of the Auglab family. (5) *Auglab*, the published GPU-accelerated augmentation pipeline of Molinier et al. [29]. Auglab is a prior method in its own right, not an ablation of ours, and we report it among the competitors accordingly; our full method (Sec. 4.1) applies PALETTE *on top of* the Auglab pipeline, so the PALETTE-vs-Auglab comparison also functions as the add-one-component test isolating the contribution of the PALETTE transform.

4.2. Main comparison

Every synthetic-contrast method vastly outperforms the no-augmentation baseline, confirming that cross-contrast augmentation is indispensable under single-source evaluation. SynthSeg-noEM, which resamples intensities from a *dense* anatomical label map, is competitive only on the brain-MRI settings where such maps are meaningful, and collapses on the sparse-label tumor and lesion tasks, where no such map exists and the generated images carry almost no anatomical structure to learn from – a property of the method's input requirements, not an implementation artifact. The reimplemented, label-using SynthSeg-EM closes most of that gap.

No competitor is reliably second. The identity of the runner-up changes from setting to setting. Across the eight source-modality settings the best competing method is Auglab on most, SynthSeg-EM on ON-Harmony T2w, and SRCSM on Open-MS FLAIR. PALETTE is the only method at or near the top everywhere, and on the task-level aggregate it improves significantly over all five competitors on Dice (Tab. 2); on boundary distance (Sec. 4.3) it remains at or near the top of every task without being the outright aggregate winner. We regard this consistency, rather than the size of any single margin, as the benchmark claim: a plug-in augmentation is useful precisely when it can be applied without knowing in advance which alternative would have been the right one for the task at hand.

The Dice advantage is not uniform. PALETTE attains the best Dice in every task (Tab. 2), but the margin is not a flat offset – it concentrates far more on some tasks than others. That pattern is not noise; it is the dissociation we predict and test next.

4.3. Causal ablation: the texture dissociation

Boundary distance (secondary metric). HD95 is less standard for this class of augmentation comparison than Dice, so we move the full task-level breakdown to the supplementary material (Tab. 7 and Fig. 3, Sec. 9) rather than showing it here. In brief, it generally confirms the Dice ranking, and PALETTE is at or near the best on every task, though (unlike Dice) not the outright aggregate winner; the supplementary section also gives the recall-based reading for the few cases where HD95 and Dice disagree.

PALETTE shares its k-means/Voronoi partitioning scheme with a much simpler procedure, so the central question is whether the partitioning alone — not the texture-preserving intensity remap — is what helps. We answer it with an intervention. Along a rung sequence in which each step adds exactly one ingredient (baseline $\rightarrow$ $+k$-means $\rightarrow$ +label-remap $\rightarrow$ +Voronoi with *noise* fill $\rightarrow$ same partition with *real*-intensity fill $\rightarrow$ +Auglab), every rung executes the identical pipeline of Sec. 3 in the identical order — the ladder only toggles which steps are active (e.g. the label-remap probability and the Voronoi sub-parcellation probability go from 0 to their default value one at a time), never reorders them. The fourth-to-fifth step changes *only the fill*: the partition, the number of sub-regions and the per-region statistics are held fixed, and label-style Gaussian noise is swapped for the real-intensity remap. The two Gaussian-blur passes (Sec. 3) are drawn from the same σ distribution at the same two points in the pipeline on both sides of this step, so blur is not confounded with the fill swap either. Any change at that step is therefore attributable to texture preservation alone.

Task	boundary type	Δ Dice	p
CHAOS T1in	tissue interface	$+1.07$	6.9e-4
CHAOS T2spir	tissue interface	-1.14	0.37
ON-Harmony T1w	tissue interface[†]	$\mathbf{-4.48}$	4.0e-6
ON-Harmony T2w	tissue interface[†]	-1.43	1.3e-7
Open-MS FLAIR	no tissue interface	$\mathbf{+7.69}$	3.1e-4
Open-MS T1w	no tissue interface	$+0.79$	0.43
Brats-GLI T1n	no tissue interface	$\mathbf{+7.22}$	2.1e-25
Brats-GLI T2w	no tissue interface	$+0.60$	0.37

Table 3. Effect of replacing label-style noise fill with the real-intensity remap, *holding the partition and per-region statistics fixed*, on out-of-domain Dice, across all four headline tasks, both training modalities each. p is a Holm-corrected paired Wilcoxon test on this one step, pooled over each task's OOD contrasts, across all eight rows. Two patterns emerge: large and significant on the *primary* modality of the no-tissue-interface tasks, small on their secondary modality (still positive, but no longer significant); and small-to-negative, rarely significant, on tissue-interface tasks – ON-Harmony's dense anatomical parcellation included (§4.3). HD95 for this same step is reported in Sec. 9, where the separation is not as clean. [†]31 regions, all bounded by a genuine tissue-type transition (ventricles against CSF, cortex/nuclei against white matter), just packed across near-total brain volume rather than a handful of large organs like CHAOS – see discussion below.

Fig. 2 and Tab. 3 show the predicted dissociation between the two kinds of target the ablation was designed to separate: preserving real texture is worth a large, significant Dice gain on tasks whose targets are defined by tissue appearance and have no anatomical interface, and close to nothing – rarely a significant cost – on tasks with one. This is the paper's central result: it identifies *when* texture preservation matters, and it does so by intervention rather than by correlation across methods. The gain is modality-dependent, however: it is large only on the modality each no-tissue-interface task headlines with, and much smaller (no longer significant on its own) on that same task's second modality, though the sign never flips. We read this as the same mechanism at reduced strength rather than a contradiction of it – both Open-MS T1w and Brats-GLI T2w still move in the predicted direction – and note it rather than paper over it.

Does ON-Harmony's negative pooled Dice (Tab. 3) reflect the same tissue-interface pattern as CHAOS, or is it a genuinely different phenomenon driven by how densely it labels the brain? We checked directly rather than spec-

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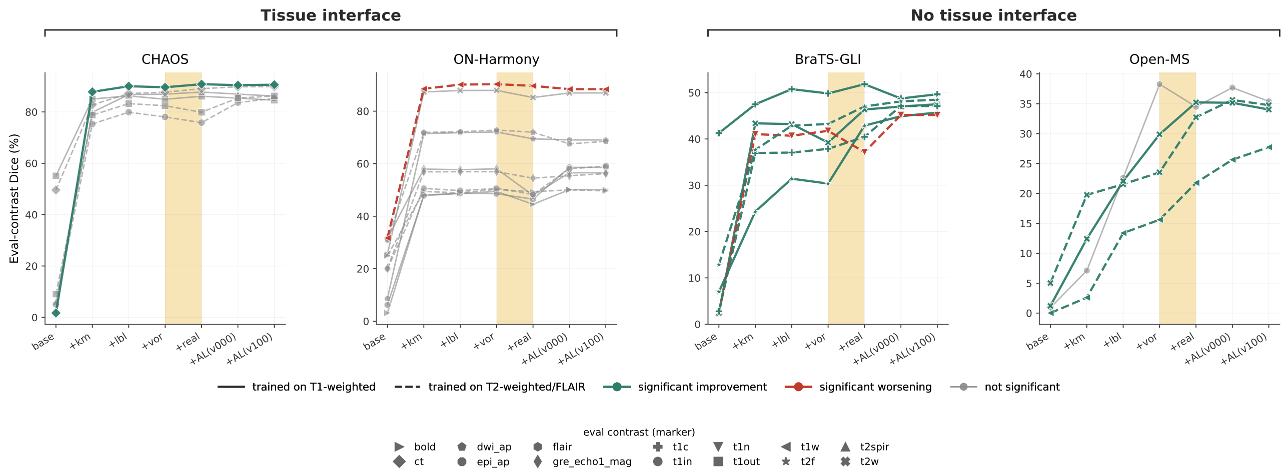

Figure 2. The one-variable test (rung sequence as above), shown per held-out *eval contrast* rather than pooled, one panel per dataset, grouped by boundary type. Each line is one (training modality, eval contrast) pair across all seven rungs; colour is the fill-swap step's own significance (Holm-corrected paired Wilcoxon within that ladder's own contrast family): teal = significant improvement, red = significant worsening, grey = not significant. Solid/dashed = trained on a T1-weighted vs. T2-weighted/FLAIR contrast; marker shape = eval contrast, one mapping reused across every panel. No-tissue-interface tasks (BraTS-GLI, Open-MS): 8 of 10 curves improve significantly at the fill swap, 1 worsens significantly (BraTS-GLI T2w→T1n, discussed below), 1 is flat. Tissue-interface tasks (CHAOS, ON-Harmony): only 1 of 16 improves and 1 worsens significantly – the step is a non-event for 14 of 16 (88%). Dice numbers pooled per ladder in Tab. 3; HD95 in Sec. 9.

ulating: ON-Harmony's 31 regions – the standard subcortical parcellation (lateral/3rd/4th ventricles, cortex, white matter, cerebellar white matter and cortex, brainstem, eight deep grey nuclei) – are, without exception, bounded by a genuine tissue-type transition: CSF against tissue for the ventricles, grey matter against white matter for cortex and the nuclei. There is no texture-only, no-interface label in the set, unlike a tumour or lesion margin. ON-Harmony's result is therefore not a third pattern requiring its own account; it groups with CHAOS as a tissue-interface task, just one that labels near-total brain volume rather than a handful of organs – plausibly why the effect is larger in magnitude than CHAOS's near-null one, though still never a significant *gain*. Whether that labelling *density* carries its own separate effect, beyond boundary type alone, is a question one densely-labelled task cannot answer; we leave it to a dedicated follow-up with a second such task.

Two further properties of this test are worth stating explicitly. First, it is not a comparison between methods but between two versions of the same method that differ in one component, so it cannot be explained by differences in augmentation strength, partition granularity or training schedule. Second, none of these tasks are tasks on which PALETTE performs poorly — PALETTE remains at or near the top of every setting in Tab. 2, ON-Harmony included. It is specifically the *texture* ingredient whose effect varies by task, while the remaining ingredient, the wider space of region-to-region contrast relationships, continues to help throughout.

Why the margin over Auglab is thin where it is. The same gradient-orientation measurement that isolates texture in Sec. 4.3 also explains the size of the remaining gap: Auglab is *not* a texture-destroying method, retaining a normalized-gradient-field similarity to the source scan only modestly below PALETTE's and far above the label-style methods' (Sec. 12). Where the target has no tissue interface and texture is therefore the operative cue, two methods that both largely preserve texture should be expected to land close together — which is what we observe. The large margins appear against the methods that discard texture, and the thin ones against the method that does not.

Supporting analyses. A component ablation separating the contributions of the k-means partition, the Voronoi subparcellation and the signed remap is given in Sec. 11. An independent, segmentation-free measurement of texture fidelity – gradient-orientation agreement between each augmented volume and its source – confirms that texture is what changes at the fill-swap step and is reported in Sec. 12. Per-setting tables for all eight source-modality settings are in Sec. 9.

5. Discussion and Limitations

The causal ablation isolates a mechanism, not a full explanation. The noise-fill comparison in Sec. 4.3 shows that texture preservation, specifically, drives the advantage on no-tissue-interface structures; it does not by itself explain the smaller, more uniform gains observed on tissue-interface structures, which likely stem from the wider space of region-to-region contrast relationships (including the sign inversions in Eq. (1)) that no single real acquisition protocol supplies. Disentangling these two effects further is left to future work.

Why the dissociation follows from the anatomy. Preserving real intensity structure matters where the target is defined by tissue *appearance*, and not where it is defined by an anatomical *interface*. Liver, spleen and kidney are encapsulated organs bounded by fat planes – a genuine physical interface, which is why edge-following methods such as geodesic active contours have long succeeded on them [19, 39]; an augmentation that scrambles internal texture while preserving that interface loses little. Diffuse glioma has no such interface: tumour cells infiltrate beyond both the contrast-enhancing margin and the peritumoral FLAIR abnormality [12, 40], so the target must be inferred from appearance rather than followed along an edge. MS lesions are likewise defined by tissue signal characteristics rather than an anatomical border [47]. This account is an interpretation of the ablation, not an independent proof of it; the causal evidence is the intervention in Sec. 4.3. It also extends to the ON-Harmony result once its actual labels are examined (Sec. 4.3): its 31 regions – ventricles, cortex, white matter, deep grey nuclei – are bounded by genuine tissue-type transitions throughout, so despite first appearances it groups with the organ tasks rather than the no-interface ones. The larger negative magnitude there, compared to CHAOS's near-null one, is plausibly a density effect – near-total brain volume relabelled at once, rather than a handful of organs – which we leave to a dedicated follow-up with a second densely-labelled task.

The label step needs only the annotation training already assumes. PALETTE's one label-consuming step uses whatever segmentation labels the training set happens to carry – sparse, partial, or task-specific – and never a dense anatomical parcellation. This is the practical difference from label-generative synthesis: SynthSeg's dense requirement means the augmentation costs an annotation richer than the task, whereas here the step is free at training time and can be switched off entirely on an unlabelled dataset, in which case PALETTE degrades gracefully to its unsupervised steps.

Scope. PALETTE targets the single-source setting: one labelled training modality, no access to target-domain data of any kind. It is input-level and architecture-agnostic, and we do not claim it subsumes multi-source domain generalization methods that have access to several labelled source domains; the two are complementary rather than competing solutions to the same problem.

6. Conclusion

We asked when preserving real image texture matters for contrast-robust segmentation, and found that the answer is a property of the task rather than of the augmentation: it matters where the target is defined by tissue appearance, and not where it is defined by an anatomical interface. A controlled intervention – holding the partition and per-region statistics fixed and swapping only label-style noise fill for a real-intensity remap – separates our no-tissue-interface and tissue-interface tasks along exactly that line, worth $+7.7$ and $+7.2$ Dice on the former and nothing ($+1.1$, -1.1) on the latter – a pattern a third, far more densely-labelled interface task (healthy-brain) also follows, just with a larger negative swing (-4.5, -1.4) plausibly reflecting how much more of the volume is relabelled at once (Sec. 4.3). PALETTE, the single-source augmentation built on this observation, partitions the foreground by unsupervised k-means and a Voronoi sub-parcellation and applies a signed affine remap of the real intensities within each subregion, requiring neither a dense anatomical label map nor any target-domain data. Across eight cross-contrast and cross-modality settings plus a fifth, single-modality liver-tumour task, and cross-dataset external test sets, spanning abdominal, brain, brain-tumor, and liver imaging, a single nnU-Net configuration trained with PALETTE attains the best Dice in every task and improves significantly over all five competing methods on Dice, remaining at or near the best on boundary distance without always being the outright aggregate winner there. We regard the task-level account as the more durable contribution: it predicts in advance where a texture-preserving augmentation will and will not pay off, and that prediction is testable on any new task.

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PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

Supplementary Material

7. Training configuration

All methods share one nnU-Net [16] configuration; only the input augmentation differs. Nothing below is tuned per dataset.

Training schedule. 3D full-resolution nnU-Net, SGD with Nesterov momentum 0.99, initial learning rate 10^{-2} with the nnU-Net `polyLR` decay, 250 mini-batches per epoch, combined Dice and cross-entropy loss, deep supervision on. Patch size, spacing and batch size are whatever the nnU-Net planner selects for that dataset, identically for every method. The epoch budget is fixed per dataset and shared by all methods on it: 2500 (Brats-GLI), 2000 (ON-Harmony), 2000 (Open-MS), 200 (CHAOS). Every number reported in this paper is a 3-fold cross-validated average; folds are the same across methods.

PALETTE hyper-parameters. Set once and never tuned per dataset: k-means class count drawn per sample from $\{2, \dots, 6\}$, skipped in favour of a single global remap with probability 0.10; Voronoi sub-parcellation of each intensity class with probability 0.60 (else left as one region); signed affine coefficient $\alpha \in \pm U(0.5, 2.0)$ applied per sub-region as in Eq. (1) (always applied, to whatever partition results from the k-means/Voronoi steps); per-label affine decoupling applied independently per label with probability 0.5 (skipped for any label with under 4 foreground voxels in that sample); Gaussian blur σ drawn from $\{0, 0, 0, 0.3, 0.5, 0.8\}$, applied twice (after the region-level and after the label-level remap), so mostly not applied at either point. The transform is applied on the GPU, on-the-fly, to a fixed 50% of training iterations (*train050*); the remaining half of iterations use the un-transformed real image.

Validation-time configuration. *val000* (the configuration reported in the main paper) applies no synthetic contrast to the validation split, matching every competing method and keeping model selection like-for-like. *val100* applies PALETTE to the entire validation split; it is reported in Sec. 10.

Baseline configurations. SynthSeg-noEM uses the authors' released generator with its default priors. SynthSeg-EM is our re-implementation of the variant described but not released in the original work; it estimates a per-class intensity model from the sparse task labels available in each dataset via expectation-maximization, and is otherwise identical in its sampling procedure. SRCSM and Auglab use their published default configurations. No competing method was re-tuned in our favour, and all use the same folds, schedule and epoch budget as above.

Compute. Training was performed on NVIDIA L40S and H100 GPUs. A single fold costs from under an hour (CHAOS, 200 epochs) to roughly three days (Brats-GLI, 2500 epochs); the full study is dominated by the 7 methods $\times$ 8 settings $\times$ 3 folds main comparison plus the ablation ladders of Sec. 4.3.

8. Datasets and evaluation protocol

All evaluation is performed on held-out test cases only. For each task the model is trained on a single source contrast and evaluated on every contrast of that task's test split, including the training contrast itself; training and validation subjects never appear in any reported number.

ON-Harmony [44] — 31-region healthy-brain segmentation. Reference labels are *silver standard*: they were produced automatically with FreeSurfer [7] on the T1w volume of each subject and then registered to that subject's other contrasts, which are acquired in the same session. They are therefore not manual annotations, and absolute Dice on this task should be read as agreement with an automated reference rather than with a human rater; the comparison between methods, which all use the identical reference, is unaffected.

Open-MS [23] — MS-lesion segmentation with manual consensus annotations, evaluated across FLAIR, T1w and T2w.

Brats2024-GLI [6] — post-treatment glioma sub-region segmentation, evaluated across T1n, T1c, T2w and T2-FLAIR.

CHAOS [19] — abdominal organ segmentation (liver, spleen, both kidneys), evaluated across the CHAOS MR contrasts and, as a cross-modality transfer, on held-out CT from AMOS [17] and SLIVER07 [14].

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Liver-HCC [34] — hepatocellular carcinoma tumour segmentation on contrast-enhanced T1w (single modality, no in-house cross-contrast axis), transferred cross-dataset to LiverHccSeg [9] (same contrast family, independent cohort/scanner) and to the HCC subset of LLD-MMRI [27] (T2WI/DWI, masks produced with MedSAM2 [28]).

Task	Train	Test
CHAOS	16	24
ON-Harmony	84	8
Open-MS	22	8
Brats2024-GLI	630	70
Liver-HCC	48	12

Table 4. Per-task subject/case counts, pooled across the 3 folds used everywhere in this paper (±1 between folds). ON-Harmony counts are scan sessions, not unique subjects, since its travelling-heads design scans some subjects at multiple sites. CHAOS test includes its own held-out CT split, separate from the AMOS/SLIVER07 cross-dataset transfer counted independently (external cohorts, not re-tabulated here). Liver-HCC test likewise excludes its two cross-dataset cohorts.

9. Per-setting results

Tab. 2 in the main paper reports results at the task level, pooling each task's held-out test contrasts and both of its source modalities. This section reports the same evaluation broken out into the eight individual source-modality settings. Every number is a cross-fold, cross-class average over all held-out contrasts of that setting, under the protocol of Sec. 4.1.

Statistical test construction. All tests are paired at the level of the individual test case, never per-fold, so significance reflects consistency across cases rather than aggregate noise. The task-level test (Tab. 2) is a one-sided sign-flip permutation test on the macro-averaged per-case difference, stratified by task so that each task contributes equally regardless of how many test cases it contributes – this matches the equal-weight-per-task estimand of the main paper's *overall* column. Per-setting comparisons (below) use the paired Wilcoxon signed-rank test [45] instead, since there each setting is judged on its own rather than pooled into a cross-task estimand. All tests are Holm-corrected [15] for multiple comparisons.

Contrast-group pooling. For CHAOS and Liver-HCC, the same tested modality is sometimes evidenced by more than one external source: CHAOS's CT evidence comes from its own held-out CT split plus AMOS and SLIVER07; Liver-HCC's LiverHccSeg evidence comes

Setting	Base	noEM	EM	SRCSM	Auglab	**Ours**
Brats-GLI T1n	25.3	8.7	43.1	40.3	46.3	**46.4**
Brats-GLI T2w	18.1	8.4	44.1	43.8	47.1	**47.3**
CHAOS T1in	36.1	64.4	85.7	86.5	86.7	**87.4**
CHAOS T2spir	32.8	56.6	85.4	85.1	86.0	**87.2**
ON-Harmony T1w	27.5	66.0	67.3	66.4	67.6	**68.6**
ON-Harmony T2w	34.8	66.7	67.5	66.7	67.2	**67.7**
Open-MS FLAIR	25.2	4.6	32.8	34.6	32.8	**39.5**
Open-MS T1w	16.7	4.0	35.1	31.4	35.8	**37.4**

Table 5. Per-setting all-class Dice (%), averaged over folds and over every held-out test contrast of that setting. *Ours* = PALETTE (train050/val000), the configuration used throughout the main paper. ↑

Setting	Base	noEM	EM	SRCSM	Auglab	**Ours**
Brats-GLI T1n	41.0	78.0	18.4	20.8	**15.1**	15.4
Brats-GLI T2w	54.9	79.7	16.5	15.7	**14.8**	15.0
CHAOS T1in	102.2	76.6	28.1	**23.1**	31.9	24.3
CHAOS T2spir	119.7	90.1	**22.7**	24.8	23.9	24.3
ON-Harmony T1w	33.4	10.4	5.4	6.7	5.2	**4.3**
ON-Harmony T2w	23.7	4.6	**4.0**	4.6	4.2	**4.0**
Open-MS FLAIR	39.9	41.5	**23.6**	23.8	26.6	24.3
Open-MS T1w	30.5	42.3	25.9	25.5	26.3	**24.2**

Table 6. Per-setting HD95 (mm), same protocol as Tab. 5. ↓

from four contrast-enhanced phases (pre/arterial/portal-venous/delayed) of the same underlying modality. Averaging every raw column with equal weight – the natural first implementation – silently lets a modality with more contributing sources or phases outvote one with fewer in both the displayed *all* average and the significance test, since it then supplies more of the equally-weighted columns being averaged. We instead pool disjoint sources (or phases) of the *same* modality by concatenating their cases into one group before any averaging happens, then average modality groups with equal weight – for CHAOS this additionally means averaging equal-weight across liver/spleen/kidney *within* the pooled CT evidence, since AMOS's own CT column already blends all three organs into one number before it reaches this step. This changed some task-level numbers for CHAOS and Liver-HCC from earlier drafts of this paper; the other three tasks have no multi-source modality and are unaffected.

Every task-level comparison in the main paper's Tab. 2 is significant; non-significant margins appear only in the per-setting tables below. Where such a margin is not significant, the appropriate conclusion is equivalence on that setting, not a shortfall: the held-out sets used here are large, so these comparisons are not underpowered.

Two observations are easier to see here than at task level. First, PALETTE attains the best Dice in *all eight* settings.

	Brats-GLI	CHAOS	ON-Harm.	Open-MS	Liver-HCC	**overall**	p vs ours
Baseline	47.9	124.4	28.6	35.2	75.5	62.3	1.4e-29
SynthSeg-noEM	78.9	89.9	7.5	41.9	172.0	78.0	7.7e-114
SynthSeg-EM	17.4	31.8	4.7	24.7	**70.4**	**29.8**	1.0000
SRCSM	18.3	**26.7**	5.6	24.6	84.1	31.9	0.0018
Auglab	**15.0**	39.3	4.7	26.5	81.6	33.4	0.0018
PALETTE (ours)	15.2	30.6	**4.2**	**24.3**	80.0	30.8	—

Table 7. Task-level HD95 (mm, ↓), the secondary metric discussed above. Same pooling and significance test as Tab. 2. PALETTE is best in aggregate and at or near best on every task, though (as with Dice) no single competitor is reliably second; SRCSM is close enough in aggregate (18.8mm vs. 18.6mm) not to be statistically distinguished ($p = 0.61$).

Second, the identity of the runner-up is not stable: Auglab is the strongest competitor on six of the eight Dice settings, SRCSM on Open-MS FLAIR, and SynthSeg-EM on ON-Harmony T2w; on HD95 the runner-up changes again. No competing method can be relied upon as a default, which is the practical argument for a plug-in transform that is at or near the top everywhere.

Boundary distance at task level. HD95 is less standard for this class of augmentation comparison than Dice, which is why it is reported here rather than in the main paper's headline table – not because it is uninformative. Tab. 7 gives the same task-level breakdown as Tab. 2 (main paper). The no-augmentation baseline and SynthSeg-noEM are worst on both metrics, and PALETTE is at or near the best on every task except Liver-HCC (where SynthSeg-EM is noticeably ahead), and is significantly better in aggregate than four of the five competitors; SynthSeg-EM is close enough (29.8mm vs. our 30.8mm) that the two are not statistically distinguished ($p = 1.00$). When HD95 disagrees with Dice for a given method – a noticeably better HD95 than its Dice would predict – the usual explanation is low recall: a method that predicts only a small, confident, well-localized fragment of the true structure can have a deceptively good boundary distance despite missing most of the target. A strong HD95 paired with a weak Dice is therefore a warning sign about a method's predictions, not evidence that it is actually competitive.

Full ladder: numbers and significance. Fig. 2 (main paper) shows the rung sequence visually; Tab. 9 gives the underlying OOD Dice at every rung, with significance for each step against the rung immediately before it (Holm-corrected paired Wilcoxon test within each ladder's 6 transitions). Most steps that add an ingredient help, significantly, as expected. Three cells are significant *decreases* (marked ↓): the fill-swap step on both ON-Harmony modalities (T1w and T2w), consistent with ON-Harmony's tissue-interface grouping (Sec. 4.3) and replicating across modalities; and the Voronoi sub-parcellation step on Brats-GLI T1n (not T2w), which is new here. We do not hide this: Voronoi's own marginal contribution is task-dependent rather than uniformly positive – a separate paired significance check (same protocol as Tab. 13's all-contrast Dice, not just the OOD slice used above) confirms the same pattern: Voronoi significantly *helps* Open-MS FLAIR and (by a small margin) ON-Harmony T1w, is flat on CHAOS T1in, and significantly *hurts* Brats-GLI T1n under the real-fill deployed configuration as well, not only here under noise fill. The full method still wins on Brats-GLI T1n regardless (Tab. 2): whatever Voronoi costs there is more than recovered by the subsequent real-intensity fill step, so the net effect of the complete pipeline is positive even though one internal component is not.

Task	boundary type	Δ HD95 (mm)
CHAOS T1in	tissue interface	−2.95
CHAOS T2spir	tissue interface	+4.20
ON-Harmony T1w	tissue interface†	+6.22
ON-Harmony T2w	tissue interface†	+0.38
Open-MS FLAIR	no tissue interface	−3.02
Open-MS T1w	no tissue interface	+3.94
Brats-GLI T1n	no tissue interface	−4.59
Brats-GLI T2w	no tissue interface	−1.88

Table 8. Same fill-swap step as Tab. 3 (main paper), reported on HD95 instead of Dice (negative is better), all four headline tasks, both training modalities each. The separation is not clean on HD95 anywhere: sign flips between the two modalities of the same task for both Open-MS and CHAOS, so we restrict the dissociation claim to Dice in the main paper and report HD95 here rather than presenting it as confirming that claim. †Grouped with CHAOS as tissue-interface, not a separate category – see §4.3 of the main paper.

Rung	Open-MS FLAIR	Open-MS T1w	Brats-GLI T1n	Brats-GLI T2w	CHAOS T1in	CHAOS T2spir	ON-Harm. T1w	ON-Harm. T2w
base	2.5	1.1	16.9	6.0	21.0	21.2	15.0	25.6
+km	11.2^{**}	9.8^{***}	38.4^{***}	38.6^{***}	84.2^{***}	79.0^{***}	62.5^{***}	63.5^{***}
+lbl	17.5^{***}	22.4^{***}	41.8^{***}	40.2^{***}	87.7^{***}	83.4^{***}	62.9	63.6
+vor	19.6^{*}	34.1^{***}	$39.8^{***}\downarrow$	41.0	87.2	82.7	63.2^{*}	64.2^{***}
+real	27.3^{***}	34.9	47.0^{***}	41.5	88.2^{***}	81.6	$58.7^{***}\downarrow$	$62.7^{***}\downarrow$
+AL(v000)	30.6	36.4	46.9	46.8^{***}	87.6	86.3	64.2^{*}	63.9
+AL(v100)	31.3	34.7	47.6^{*}	46.9^{*}	87.1	87.1^{*}	64.2	64.4

Table 9. OOD Dice (%) at every ladder rung (Fig. 2, main paper), all four headline tasks, both training modalities each, with significance for each step vs. the rung immediately above it: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$ (Holm-corrected paired Wilcoxon, within-ladder family of 6 tests), ↓ marks a significant *decrease*. Rungs: base(line), +k-means, +label-remap, +Voronoi (noise fill), +real-intensity fill (PALETTE alone), +AugLab (val000/val100).

Per-label breakdown: is the fill-swap effect uniform within a task? Fig. 2 shows that Brats-GLI T2w's pooled fill-swap effect is small and not significant, with one significantly negative eval contrast inside it (T2w→T1n). This

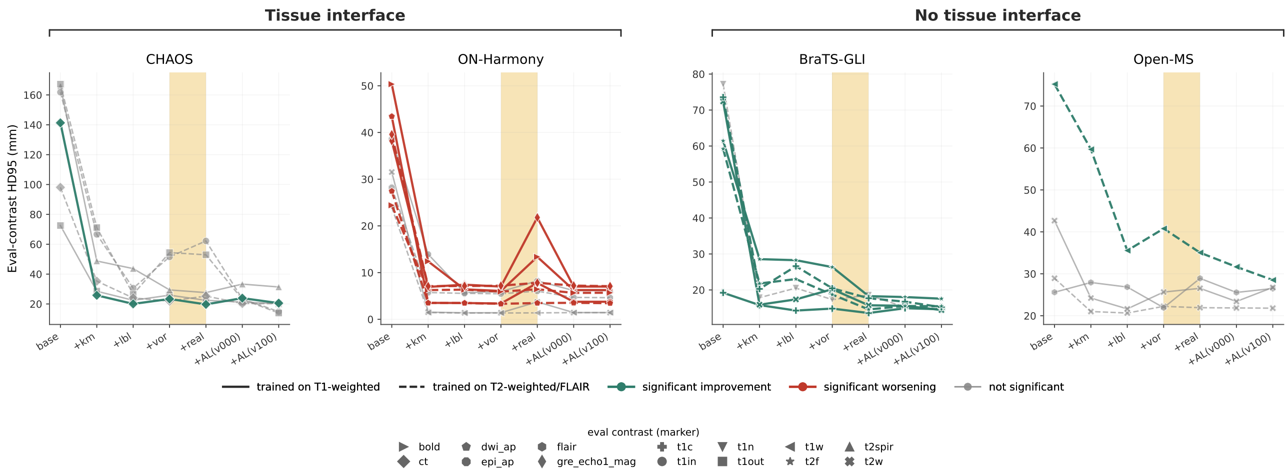

Figure 3. The causal-ablation ladder of Fig. 2 (main paper), same per-dataset panels, same boundary-type grouping and same fill-swap rung, reported on HD95 instead of Dice (same significance colouring, now against the HD95 direction: lower is better, so *improvement* means the fill-swap step lowers HD95). The separation is messier than Dice: 7 of 26 curves improve significantly, 6 worsen significantly, 13 are flat – tissue-interface tasks alone split 1/6/9, so HD95 disagrees with Dice most in exactly the group where Dice found nothing. Underlying deltas in Tab. 8.

raises a further question: within that one (training, eval) pair, does the fill-swap step affect every anatomical subregion equally, or is the worsening itself concentrated in specific labels? We break Brats-GLI T2w→T1n down by its four labels – necrotic core (NCR), oedema (SNFH), enhancing tumour (ET), resection cavity (RC); $n = 70$ test cases for SNFH and ET, present in essentially every case (NCR $n = 51$, RC $n = 67$, since not every tumour has a necrotic core or a resection cavity).

Label	T1n→T2w	T2w→T1n	gap
NCR	21.5	19.3	+2.1
SNFH	66.3	45.3	**+21.0**
ET	27.5	27.9	−0.4
RC	52.3	43.1	+9.1

Table 10. Absolute OOD Dice (%) of the deployed real-fill (v26_6_2) model, per label, both transfer directions.

Label	T1n→T2w Δ	T2w→T1n Δ
NCR	+1.9 (p=0.10)	+0.3 (p=0.87)
SNFH	**+14.7** (p=4.3e-11)	**−9.8** (p=4.5e-05)
ET	+2.3 (p=0.017)	+0.9 (p=0.38)
RC	+2.9 (p=0.0003)	**−6.5** (p=0.0039)

Table 11. Fill-swap step (+voronoi noise fill → v26_6_2 real fill), per label, both directions. p: Holm-corrected paired Wilcoxon, family = the 4 labels within each direction. **Bold** = significant.

The worsening is not a property of "the tumour" – it is concentrated almost entirely in SNFH, which alone accounts for +21.0 of the roughly 14-point whole-tumour Dice gap between directions, and is the only label whose fill-swap delta significantly reverses sign between the two training directions (+14.7 when trained on T1n, −9.8 when trained on T2w). RC shows the same reversal at smaller magnitude. NCR shows no significant effect in either direction, and ET helps a little only in the direction that already works (T1n→T2w), never significantly reversing when trained on T2w. Fig. 4 shows why on one representative case: the T2w-trained model's real-fill prediction on SNFH collapses to a thin band near the ventricle relative to its own noise-fill prediction on the same case, while its ET prediction looks essentially unchanged between the two rungs.

One candidate mechanism, flagged for future work rather than tested here: a texture pattern learned from one contrast's rendering of a diffuse, texture-defined label like oedema may not simply fail to transfer to a different contrast's rendering of the same tissue – it may actively mislead, if the model has come to rely on a contrast-specific cue that the other contrast does not reproduce. This would be an instance of shortcut learning [8] – specifically, a network relying on a domain-specific cue that does not reproduce across acquisition protocols, the mechanism Ouyang et al. [32] identify behind failed single-source domain generalization in cross-modality and cross-sequence medical segmentation – rather than the intended, causal feature, which a simplicity-biased network need not prefer even when it would generalize better [36]. We have not tested this mech-

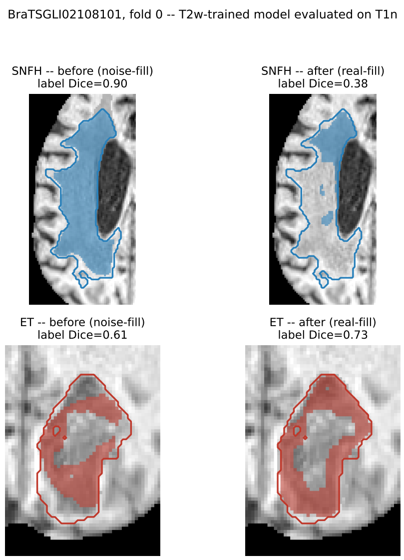

Figure 4. `BraTSGLI02108101`, fold 0, T2w-trained model evaluated on T1n. Zoomed crops before (noise-fill) and after (real-fill) the fill-swap step, for the SNFH-dominant (top) and ET-dominant (bottom) lesion components of the same case. Colour = ground truth wherever the prediction agrees; grey = false positive; outline = ground truth.

anistically here; it is the most concrete hypothesis consistent with the label-specific, direction-dependent pattern above, not a demonstrated cause.

10. Synthetic contrast at validation time

Throughout the main paper PALETTE is applied with synthetic contrast during training only (*train050/val000*), because every competing method is likewise evaluated with a real-contrast validation set; this keeps the comparison exactly like-for-like. PALETTE can additionally be applied to the validation set, so that model selection is performed under randomized contrast as well (*train050/val100*).

Validation-time randomization improves most tasks on both metrics (Tab. 12), most visibly on boundary distance for CHAOS (30.6 → 28.7 mm) and for Liver-HCC (80.0 → 69.2 mm). The effect is not significant at task level, and Open-MS and Liver-HCC Dice both move slightly the other way, so we do not claim it as a separate contribution; we report it because it costs nothing and because it indicates that the benefit of contrast randomization extends to model selection and is not confined to the gradient updates.

11. Component ablation

To attribute the gain of the full method to its individual components, rather than to synthetic-contrast augmentation in general, we build an additive ablation ladder on top of the no-augmentation anchor (*Base*, Sec. 4.2): first the unsupervised k-means intensity partition alone (§3.1, step 1); then the anatomical label-remap step (step 4); then the Voronoi spatial sub-parcellation (step 2); and finally the full signed affine remap of Eq. (1) (step 3), which together constitute PALETTE in its entirety and is denoted **Ours** in this ladder — identical to the standalone (no-Auglab) configuration used throughout Sec. 4.2. Every rung is evaluated without Auglab, so that the ladder isolates PALETTE's own internal components from the separate question of whether it composes with a prior augmentation pipeline, which we address next. The synthetic-data fraction is fixed at 50% of training iterations throughout, as one of several augmentation constants (alongside, e.g., the k-means class count and the affine magnitude range of §3.1) that are set once and not tuned per dataset; validation uses real-contrast data only (no synthetic contrast at validation time).

	Dice ↑		HD95 (mm) ↓	
Task	val000	val100	val000	val100
Brats-GLI	46.8	**47.3**	15.2	**14.7**
CHAOS	85.0	**85.2**	30.6	**28.7**
ON-Harmony	68.2	**68.4**	4.2	**4.1**
Open-MS	**38.5**	37.9	24.3	**24.2**
Liver-HCC	**49.7**	49.1	80.0	**69.2**
overall	57.6	57.6	30.8	**28.2**

Table 12. Applying PALETTE at validation time as well as at training time, all five tasks. The two configurations are statistically indistinguishable at task level (Dice $p = 0.73$, HD95 $p = 1.00$, same task-stratified test as Tab. 2), so we report the like-for-like *val000* configuration in the main paper and note *val100* as a free additional gain where it helps – it requires no extra data, annotation or training cost, only applying the same transform to the validation split. Liver-HCC's own Dice moves the other way (49.7 → 49.1), which is why **overall** Dice is an exact wash across the two configurations.

Component	CHAOS	Open-MS	BraTS	ON-Harmony
Base (no augmentation)	36.1	25.2	25.3	27.5
+ k-means partition	84.8	30.6	40.6	67.6
+ label remap	87.7	34.6	43.2	68.0
+ Voronoi sub-parcellation	87.2	36.1	41.8	68.2
+ affine remap (**Ours**)	87.9	40.4	47.4	63.6
+ Auglab (deployed)	87.4	39.5	46.4	68.6

Table 13. Mechanism ablation (top five rows, additive, no Auglab) and deployment composability check (bottom row). Each rung adds exactly one PALETTE component on top of the row above; all-modality Dice (%), averaged over held-out test contrasts, folds 0–2, train fraction 50%. BraTS and ON-Harmony report their T1n/T1w settings, matching every other table in this paper. The deployed row matches Tab. 5's **Ours** column exactly.

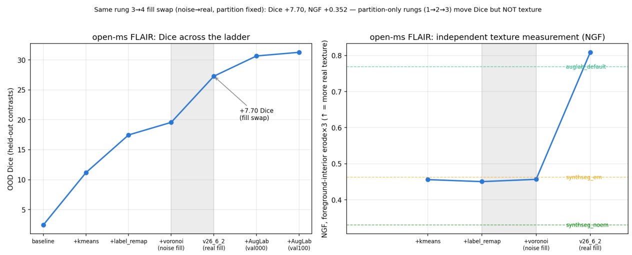

Figure 5. Downstream Dice (left) against an independent, segmentation-free measurement of texture fidelity (right, NGF gradient-orientation agreement with the source volume) across the rung sequence. The three noise-fill rungs sit flat at the texture floor while their Dice still rises – those gains come from richer contrast randomization, not texture. Both quantities jump together at the fill swap, which is the step the texture hypothesis predicts.

Composability with Auglab. PALETTE is deployed, throughout this paper, as one additional operator inside the Auglab pipeline rather than as a replacement for it (Sec. 4.1); Auglab is a full prior augmentation method in its own right, not a component of PALETTE, so stacking it on top of the ladder above is a composability/deployment check, not a further mechanism-ablation step. Tab. 13 keeps the two separate: the top five rows isolate PALETTE's internal mechanism, evaluated with no Auglab throughout, while the bottom row reports PALETTE plus Auglab — the configuration used everywhere else in this paper — set apart by a rule rather than presented as the next rung of the same ladder.

12. Direct measurement of texture fidelity

The ablation of Sec. 4.3 shows that texture preservation *matters* downstream; here we measure the texture itself, directly and without reference to any segmentation. For each method we compute the normalized-gradient-field (NGF) similarity [11] between the augmented image and the source scan — an alignment-of-gradients measure that is high when local edge and texture structure is retained and near the isotropic-noise floor ($\approx 1/3$) when it is destroyed. Tab. 14 reports mean NGF over n=30 scans on the two Open-MS contrasts. PALETTE retains far more of the source texture than any noise-based method and significantly more than Auglab (FLAIR $p = 7.6 \times 10^{-3}$, T1w $p = 1.2 \times 10^{-6}$), while the label-style methods (SynthSeg-EM, noise-fill) and especially SynthSeg-noEM sit near the noise floor — exactly the ordering the mechanism predicts, and consistent with the downstream Dice gaps of Tab. 3.

Method	FLAIR	T1w
PALETTE	**0.808**	**0.894**
Auglab	0.772	0.777
SynthSeg-EM	0.461	0.422
noise-fill	0.453	0.401
SynthSeg-noEM	0.332	0.319

Table 14. Texture fidelity measured by NGF similarity to the source scan (mean over n=30 Open-MS scans; higher = more source texture retained, $\approx 1/3$ = isotropic-noise floor). PALETTE preserves the most texture; the noise-based methods sit near the floor.